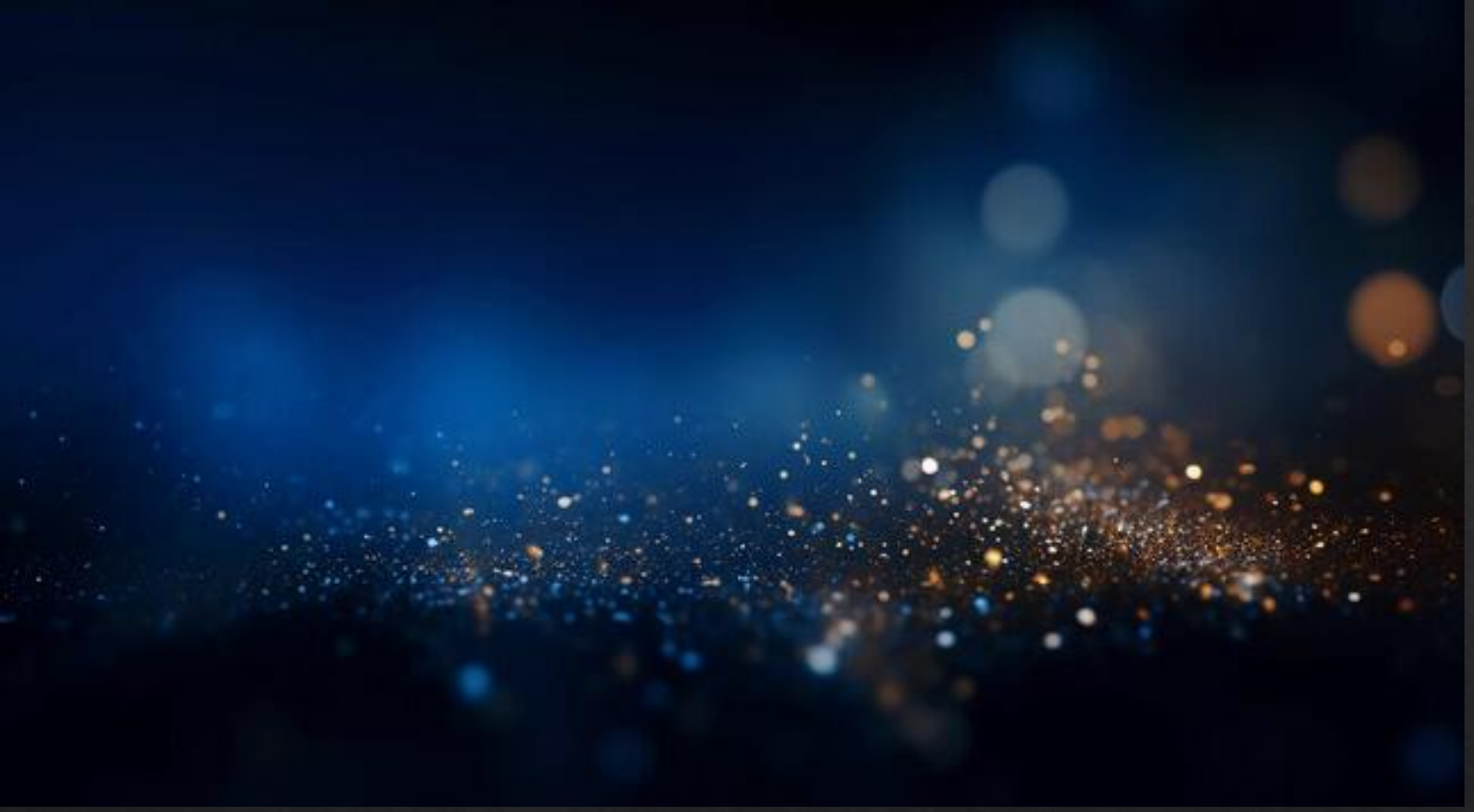


MA416

Lecture 22 – November 18, 2025

Overview of Multiple Linear Regression

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The Basics of Multiple Linear Regression

```
mydata = read.csv('CompanyPayroll_9.25.csv')
source('mymlr.R') # you should have your own by now

Y = mydata$Prior_Experience

X1 = mydata$Age
X2 = mydata$Efficiency_Factor_Obs
X3 = mydata$Hourly_Wage
X4 = model.matrix(~ mydata$Job_Rank)[,-1,drop=F]
X5 = model.matrix(~ mydata$Biological_Sex)[,-1,drop=F]
# X5 = ifelse(mydata$Biological_Sex == 'Male', 1, 0)

X = cbind(X1, X2, X3, X4, X5)
```

```
> M$SSE
[1] 23.83672
> M$RMSE
[1] 0.5424763
> M$pval
[1] 0.004064484
> M$r2
[1] 0.2219602
> M$r2adj
[1] 0.1588758
```

```
> M = mymlr(X, Y)
> rbind(c(M$bhat), c(M$sebhat), c(M$bhattstat))
                      X1         X2         X3 mydata$Job_RankAst. Manager mydata$Job_RankEntry mydata$Biological_SexMale
[1,] 0.2088406 0.04676621 0.03948700 -0.01318625                -0.3247913          -0.1189300                0.1337547
[2,] 0.4529093 0.01240059 0.03997973  0.01205473                 0.2265316           0.1530451                0.1303330
[3,] 0.4611092 3.77128767 0.98767541 -1.09386555                -1.4337572          -0.7770911                1.0262533
```

```
> Ryandx = round(cor(cbind(Y, X)),2)
> Ryandx
                              Y    X1    X2    X3 mydata$Job_RankAst. Manager mydata$Job_RankEntry mydata$Biological_SexMale
Y                          1.00  0.40  0.12 -0.08                       -0.06                -0.05                      0.17
X1                         0.40  1.00  0.25  0.13                        0.19                -0.13                      0.11
X2                         0.12  0.25  1.00  0.34                        0.49                -0.22                      0.21
X3                        -0.08  0.13  0.34  1.00                        0.39                -0.17                      0.08
mydata$Job_RankAst. Manager -0.06  0.19  0.49  0.39                        1.00                -0.55                      0.03
mydata$Job_RankEntry       -0.05 -0.13 -0.22 -0.17                       -0.55                 1.00                     -0.06
mydata$Biological_SexMale   0.17  0.11  0.21  0.08                        0.03                -0.06                      1.00
```

The Basics of Multiple Linear Regression

Theorem 1: Suppose that R_X is the correlation matrix associated to the predictors (only) for a regression model, and assume that V_k are the associated variance inflation factors of the predictors, satisfying:

$$\mathbb{V}(\hat{\beta}_k) = \frac{\sigma^2}{\sigma_{X_k}^2(n-1)} \cdot V_k = \frac{\sigma^2}{\sigma_{X_k}^2(n-1)} \cdot \frac{1}{1-\tilde{\rho}_k^2}$$

where $\tilde{\rho}_k^2 = r^2\{X_k \sim X_k^c\}$. Then $V_k = \text{diag}_k(R_X^{-1})$.

```
> round(rbind(c(M$vif), c(M$mci)), 2)
       X1   X2   X3 mydata$Job_RankAst. Manager mydata$Job_RankEntry mydata$Biological_SexMale
[1,] 1.08 1.47 1.23                 1.95                 1.45                          1.07
[2,] 1.04 1.21 1.11                 1.40                 1.20                          1.03
```

Theorem 2: If all variance inflation factors are equal to 1*, then the gradients for the predictors, X_k, for predicting the response variable, Y, can be found in the same manner as they were generated for simple linear regression. That is:

$$\hat{\beta}_k =^* (r_{x_k,y}) \cdot \frac{s_y}{s_{x_y}}$$

The Basics of Multiple Linear Regression

Note if that if all (4) of the assumptions for multiple linear regression appropriateness are met, then the standardized residuals should follow a Z distribution (be able to prove this). Hence analyzing them can be one way of assessing (some of) the assumptions. Recall that:

$$\mathbb{V}(\hat{\varepsilon}) = \sigma^2 \cdot (I - H) \quad \text{where} \quad H = X \cdot (X^T X)^{-1} X^T$$

Since $\mathbb{E}(\hat{\varepsilon}) = 0$, then $\hat{\varepsilon}_k^S = \frac{\hat{\varepsilon}_k - 0}{\sqrt{\sigma^2} \cdot \sqrt{1 - l_k}}$.

```
hist(M$sres, col='peachpuff',
     main='Standardized Residuals', xlab='sres')
```

```
source('mynormtest.R')

Ntest = mynormtest(M$sres, 0, 1)
c(Ntest$KS, Ntest$CVM, Ntest$AD)
```

```
> c(Ntest$KS, Ntest$CVM, Ntest$AD)
[1] 0.05286893 0.04807482 0.29724410
```

